

Thymosin Alpha-1

Thymic Immunomodulator | 5mg & 10mg x 10 Vials

Immune Activation | Antiviral | Anti-Cancer | SubQ Injection

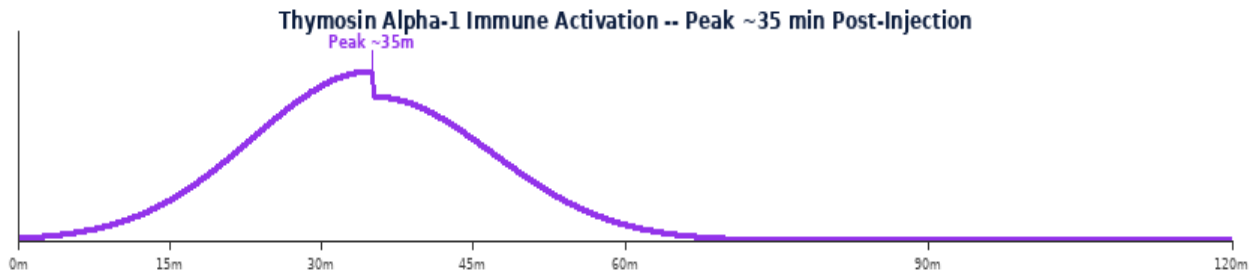
Purity: 99.8% | Endotoxin-Free | Research Use Only

Thymosin Alpha-1 (Ta1, Thymalfasin, Zadaxin) is a 28-amino acid peptide originally isolated from thymic tissue. It is a potent immunomodulator that enhances T-cell function and innate immunity. FDA-approved (as Zadaxin) in multiple countries for hepatitis B, hepatitis C, and cancer immunotherapy. Standard clinical dose is 900 mcg (0.9mg) twice weekly SubQ.



Parameter	Specification
Class	Thymic peptide immunomodulator -- T-cell activator, antiviral, anti-cancer
Vial Sizes	5mg x 10 10mg x 10 Purity 99.8%
Reconstitution	1 mL BAC Water: 5mg->5mg/mL 10mg->10mg/mL
Clinical Dose	900 mcg (0.9mg) twice weekly SubQ -- approved clinical dose
Research Dose	500-3,600 mcg SubQ no fasting required
FDA Status	Approved as Zadaxin in 35+ countries for hepatitis B/C and adjuvant cancer therapy
Cycle & Storage	12+ week courses 2-8 deg C use within 28 days

MECHANISM OF ACTION & RESEARCH BENEFITS



Pathway	Mechanism	Benefit
T-Cell Activation	Induces T-cell maturation in thymus; activates Th1 CD4+ and CD8+ cytotoxic T-cells; upregulates MHC class II expression on antigen-presenting cells; enhances T-cell receptor expression and signalling	Immune restoration; T-cell research; immunosenescence
Innate Immunity & Antiviral	Activates NK cells and dendritic cells; induces interferon-gamma and IL-12; antiviral via innate immune pathway upregulation; clinical evidence in hepatitis B/C viral clearance	Antiviral research; hepatitis; innate immunity
Anti-Cancer Adjuvant	Enhances T-cell cytotoxicity against tumour cells; restores immune function in immunocompromised cancer patients; synergistic with checkpoint inhibitors in research; approved adjuvant in cancer immunotherapy	Cancer immunotherapy; tumour surveillance; checkpoint synergy

Benefit	Context
Clinical Validation	Thymosin Alpha-1 is approved in 35+ countries as Zadaxin -- the most clinically validated immunomodulatory peptide available.
Immune Restoration	Particularly valuable in immunocompromised states, immunosenescence (aging immune decline), and post-viral immune dysfunction research.
Standard Dose	Clinical dose (900 mcg twice weekly) is well established. 5mg vial provides ~5 clinical doses. 10mg provides ~11 doses.

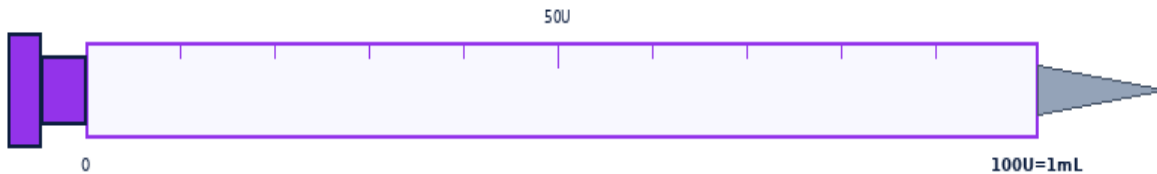
RECONSTITUTION & DOSE CALCULATION

All vials use 1 mL BAC Water. Room temperature. Swirl gently -- never shake.



1 mL BAC Water per vial. Roll gently -- NEVER shake.

Step	Action	Detail
1	Gather	Vial, BAC Water, insulin syringe (29-31G), swabs, sharps.
2	Clean stoppers	Wipe both stoppers. Air-dry 10-15 sec.
3	Draw 1mL BAC	Exactly 1 mL. Expel air bubbles.
4	Inject into vial	Aim at GLASS WALL. Slowly. Never onto powder.
5	Swirl & label	Roll 20-30 sec. Clear within 60 sec. Never shake. Label date. 2-8 deg C. Use within 28 days.



1mL Insulin Syringe | 29-31G | SubQ

Formula: $\text{Vol(mL)} = \frac{\text{Dose(mcg)}}{\text{Conc(mcg/mL)}}$ | **Units:** $\text{Vol} \times 100$

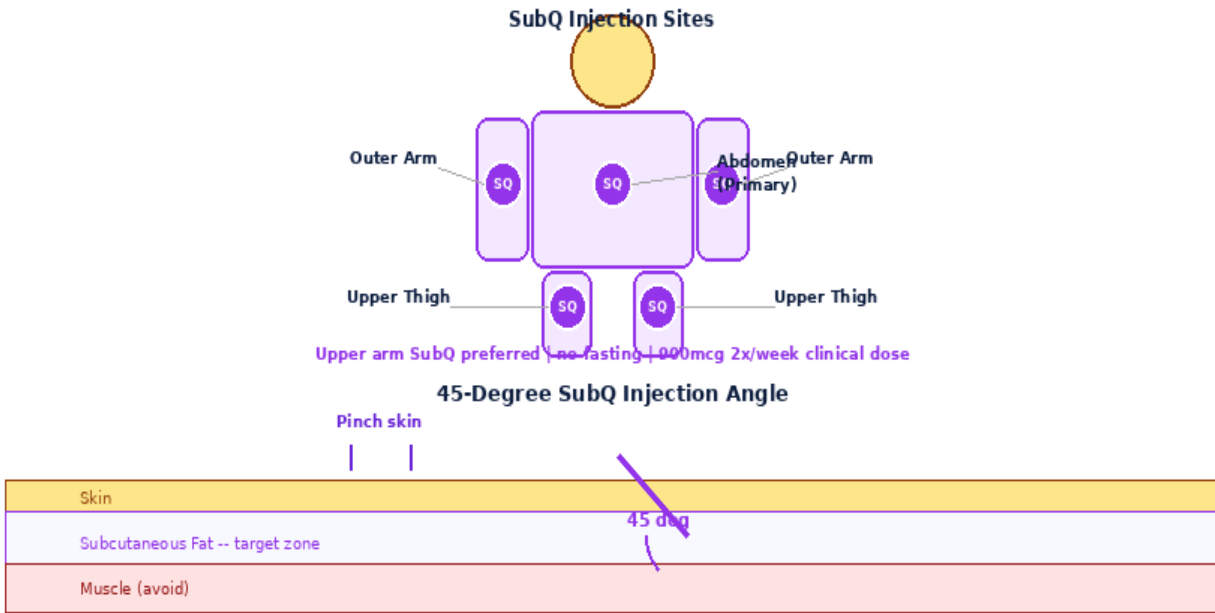
5mg x 10 -- 1mL BAC -> 5.0mg/mL (5,000mcg/mL)

Dose(mcg)	500mcg	900mcg	1800mcg	3600mcg
Vol(mL)	0.1000	0.1800	0.3600	0.7200
Units	10.0U	18.0U	36.0U	72.0U
Doses/Vial	10	5	2	1

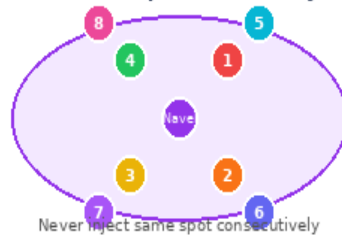
10mg x 10 -- 1mL BAC -> 10.0mg/mL (10,000mcg/mL)

Dose(mcg)	500mcg	900mcg	1800mcg	3600mcg
Vol(mL)	0.0500	0.0900	0.1800	0.3600
Units	5.0U	9.0U	18.0U	36.0U
Doses/Vial	20	11	5	2

INJECTION TECHNIQUE, PROTOCOLS & GOLDEN RULES



8-Point Rotation Map -- Rotate Every Injection



Step	Action	Detail
1	Wash & prepare	Wash hands. 29-31G insulin syringe. No fasting required.
2	Draw dose	Clinical dose: 900mcg = 0.18mL from 5mg/mL vial. Draw exact volume.
3	Inject SubQ	Upper arm preferred (clinical standard). Abdomen acceptable. 45 deg.
4	Rotate & dispose	Alternate injection sites. Sharps container.

Protocol	Dose	Frequency	Duration	Notes
Clinical (Approved)	900 mcg (0.9mg)	2x/week SubQ	12+ weeks	Approved dose for hepatitis and adjuvant cancer.
Research High-Dose	1,800-3,600 mcg	2-3x/week	8-12 wk	Higher research dose. Monitor response.

Side Effect	Likelihood	Management
Injection site reaction	Common	Mild. Alternate upper arms. Fresh needle.
Fatigue (initial)	Occasional	Immune activation response. Resolves.
Fever (immune activation)	Rare	Indicates strong immune response. Reduce dose.

Contraindications: Cancer history, pregnancy, under 18.

1. 5mg: 1mL BAC->5mg/mL 10mg: 1mL->10mg/mL.
2. Clinical dose: 900 mcg (0.9mg) twice weekly SubQ -- from 5mg/mL: 0.18mL = 18 Units.
3. No fasting required. Inject any time.
4. Upper arm SubQ is the standard clinical injection site for Thymosin Alpha-1.
5. Approved as Zadaxin in 35+ countries -- the most clinically validated immunopeptide.
6. Minimum 12-week course for full immunological benefit.
7. Store 2-8 deg C. Use within 28 days. Never freeze.
8. Monitor for fever (immune activation). Reduce dose if significant.
9. Not for use during pregnancy or in individuals under 18 without physician guidance.

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